RAJ ARYAN

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Python Backend & AI/ML Engineer | FastAPI • RAG Systems • SQL

EDUCATION

KCC Institute of Technology & Management — Greater Noida, India

BTech in Computer Science & Engineering (AI-ML) — *Oct 2021 – June 2025*

CGPA: 7.3

TECHNICAL STACK

- **Languages**: Python, SQL, C++
- **Backend & APIs**: FastAPI, Flask, REST APIs, SQLAlchemy, Alembic
- **Data & Machine Learning**: Pandas, NumPy, Scikit-Learn, NLTK, Feature Engineering, PCA
- **Generative AI & RAG**: LangChain, FAISS, ChromaDB, Ollama, Prompt Engineering
- **Databases**: PostgreSQL, SQLite
- **Tools & Platforms**: GitHub, IBM Cloud, Render, VS Code, n8n
- **Core Strengths**: Problem Solving, Critical Thinking, Rapid Learning

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WORK EXPERIENCE

AI-ML Intern - Omnie Solutions, Noida | Sep 2025 – Nov 2025

- Built and deployed machine learning pipelines using PCA and feature engineering to streamline analytical workflows and improve data processing efficiency.
- Developed a scalable E-commerce REST API using FastAPI and PostgreSQL, asynchronous SQLAlchemy for concurrent request handling.
- Managed database schema migrations using Alembic, ensuring reliable version control and smooth deployment cycles.

AI & Data Analytics Intern - Edunet Foundation | *Dec 2024 – Jan 2025*

- Developed an AI-powered chatbot using NLP techniques to improve automated query handling and conversational accuracy.
- Applied data analytics methods within the Skills4Future program to support Green AI and sustainable technology initiatives.

Emerging Technologies Intern - Edunet Foundation | *Jul 2024 – Aug 2024*

- Built a Cardiovascular Disease Prediction model using IBM Watson AutoAI, achieving **87.9% accuracy** after model tuning and evaluation.
- Developed enterprise chatbots using NLP and LLM frameworks on IBM Cloud to automate user interactions.

PROJECTS

DocuSenseAI v2 | *Python, LangChain, LangGraph, Ollama, FAISS, Streamlit, Pandas.*

- Designed **a privacy-first Corrective RAG system** enabling natural language querying over local files without mandatory pre-indexing.
- Optimized local LLM inference by refactoring model pipeline (Llama-3 → Phi-3), enabling smooth execution on **<4GB RAM systems.**
- Built a structured data ingestion pipeline for multi-sheet Excel/CSV files using Pandas and FAISS to improve retrieval accuracy with **source citations.**

VoxGuard AI | *Python, Groq API, Faster-Whisper, Librosa, Pyannote, ChromaDB, SQLite*

- Developed an **automated audio intelligence pipeline** to process and summarize long-form media using **map-reduce chunking** constraints.
- Built a semantic search dashboard for querying historical video content with vector embeddings.
- Implemented an acoustic "Trust Score" to flag low-quality audio segments and reduce hallucination risk in downstream AI outputs.

FastAPI E-commerce Backend | *FastAPI, PostgreSQL, SQLAlchemy, Alembic*

- Built a production-style REST API supporting product, user, and order workflows with **role-based access control (RBAC).**
- Implemented async database operations and transaction-safe checkout logic to prevent race conditions.
- Designed modular API architecture to support scalable backend development.

House Price Prediction System | *Python, Scikit-Learn, Pandas, Flask, JSON*

- Developed an **end-to-end regression pipeline** with feature engineering to predict housing prices on Kaggle datasets.
- Built a Flask web interface to serve real-time predictions from the trained model.

CERTIFICATIONS

- Getting started with Enterprise-grade AI (IBM Skillsbuild)
- Getting started with Enterprise Data Science (IBM Skillsbuild)